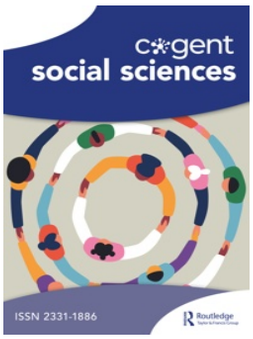

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Optimizing employee roles in the era of generative AI: a multi-criteria decision-making analysis of co-creation dynamics

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ABSTRACT
Artificial Intelligence (AI) has the ability to transform the way organizations operate, but imposes a unique challenge for them. Adoption of AI might reshape the human resource management patterns which calls for optimization of roles in workplaces for dealing with technology. Critical decision points and alternatives are identified through a systematic review of literature and Analytical Hierarchy Approach (AHP) is applied to analyze their priority weights. The results of AHP model indicate adaptive organization structure, specialized AI teams, ethics oversight and governance and innovation infrastructure to be significant in AI integration. Furthermore, sensitivity analysis technique is employed to examine the robustness of the decision outcomes at varying criteria weights and the results shows no significant impact on the ranks indicating robustness of the decision achieved if the criteria weight changes on the overall optimization process. This study contributes to expanding discussion on workforce transformation in AI era.

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Artificial intelligence; future of work; chatGPT; value co-creation; AI collaboration

SUBJECTS
Comparative Psychology; Social Psychology; Work & Organizational Psychology

Introduction

Owing to the unrelenting growth of Artificial Intelligence (AI), especially generative-AI (Gen-AI), workplaces are undergoing significant transformation (Klingbeil et al., 2024). AI tools and varied digital platforms is becoming inseparable parts of organizations over the past few years (Budhwar et al., 2023; Lazaroiu et al., 2024b). The credit goes to the superior algorithmic potential of AI in business automation, knowledge extraction and analytical, predictive and computational abilities in comparison to humans (von Krogh et al., 2023). However, assuming that the widespread adoption of AI is sufficient and alone can maintain an organization's competitiveness will be unrealistic (Lazaroiu et al., 2024a). In fact, relying completely on AI can result in homogenous work outputs that are likely to be devalued over time (Lazaroiu et al., 2023; Lin 2024). In addition, the easier it becomes to produce something, especially when humans are no longer a key part of the production, the lower the value. However, humans will never be able to follow the pace of the machines (Ozsungur, 2024). Even if the quality of work is considered a premium, the volume with which this is generated will never be on par (How et al., 2024). Neither can the human response meet the machine's ability to produce outputs on the spot. As such, managers are likely to devalue human contributions at the task level, creating a cascading effect (Lazaroiu & Rogalska, 2023). This will have repercussions for both undertaking tasks, as well as supervision and management. Human–AI collaboration is expected to be a solution to this problem (Jeremiah, 2025). Human–AI collaboration can be defined as an evolving, interactive process whereby the two parties actively and reciprocally engage in joint activities aimed at achieving one or more shared goals (Wong et al., 2021). The self-learning, natural language processing, and creative problem-solving abilities of generative-AI have the potential to profoundly reshape the dynamics of value human–AI collaboration and co-creation inside businesses and redefine employees' roles (Kiron, 2022; Muller et al., 2024; Schrage et al., 2023). For

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instance, generative AI can help process large amounts of multi-dimensional data/tasks, which cannot be handled by human cognition alone (PWC, 2022; Vallance, 2023). In the business context, this refers to a marketing manager working together with AI, namely machine learning-based systems, and complex marketing decisions can be made by bilaterally sharing information and interactive knowledge exchange. Businesses in diverse sectors frequently incorporate generative AI to enhance productivity, creativity and efficiency (DeNisi et al., 2023; Korzynski et al., 2023). This opens new possibilities for collaborative value co-creation. Co-creation, cornerstone concept for this study, is defined as a collaborative process where multiple stakeholders-including employees, AI systems, and organizational leadership-jointly contribute to creating value through shared efforts and synergies (Barile et al., 2024; Lyu et al., 2022). Generative-AI technologies such as machine and deep learning algorithms, speech recognition, computer vision techniques, natural language processing and neural networks have proven to be very powerful in credible content creation, decision-making support, and creative pursuits (Kiron, 2022; Schrage et al., 2023). This revolutionary technology adds a collaborative element in which workers and AI systems collaborate to generate value and go beyond simple automation. Conventional jobs are automized, and there is a growing possibility for human employees and AI entities to collaborate in value co-creation. While generative-AI can enhance human capabilities, the effective integration of human and AI roles remains a challenging endeavor.

When the application and capability of AI increases it tend to become complex, which in turn may pose several difficulties in optimizing employee functions (Davahli, 2020; Salleh & Aziz, 2022). The nature of employee roles is questioned by the introduction of generative-AI in the workplace. For instance, it is unclear whether Gen-AI will take away work or create more work, destroying, or creating new jobs (Budhawar et al., 2023). Employees are freed to concentrate on higher-order thinking, creativity, and strategic decision-making, where repetitive jobs become automated. As we stand at the intersection of human expertise and machine intelligence, the full utilization of generative-AI requires the strategic optimization of employee roles. This led to the formulation of our first research question, *RQ1: In the age of generative AI, what are the most important dimensions that firms take into account when rethinking employee roles to facilitate effective value co-creation?* Ensuring that employee responsibilities are not only changed but also optimized for the changing demands of the digital age requires a planned and educated decision-making process at the junction of human and machine capabilities. This leads us to the logical question *RQ2: How do various decision criteria compare to one another during the process of making decisions? RQ3: What are the various strategic trajectories or a set of choices that companies could pursue to maximize employee involvement within the framework of value co-creation and generative AI? RQ4: How can we measure and contrast the influence of various decision options?* Given the complexity of decision points, a rigorous analytical approach is required. In light of this, our research aims to explore the intricate and varied terrain of employee job optimization in the generative-AI era. The study is driven by the realization that given the complex dynamics of value co-creation, businesses' ability to succeed in this new era depends on their ability to make wise decisions about employee roles. Our goal is to use Analytical Hierarchy Approach (AHP), a Multi-Criteria Decision-Making (MCDM) approach, as a solid methodological framework to direct our investigation. Based on the research outcomes, companies can assess the degree to which AI-related choices are in line with the organization's overarching aims and strategic objectives. Preference can be given to options that strike a balance between risk reduction and innovation, and are more likely to result in the success of the organization. It is expected that corporates will benefit from this study's analysis of the requirements to implement various decisions, taking into account investments in technology, training and other pertinent aspects.

Literature review

The influence of AI adoption is of interest to both researchers and practitioners. This also includes queries and curiosities in areas such as employee displacement, skill training, changes in the nature of employment and skill composition in the workforce due to AI adoption. It is interesting to note that most research, except a few (Bankins et al., 2023; Egana-del Sol et al., 2022; Fossen & Sorgner, 2022) does not support the hypothesis of human unemployment due to AI application (Lloyd & Payne, 2019; Yang, 2022). Much research has been conducted on the impact of AI on the number and types of jobs in the

job market and changes in skill requirements (Bankins et al., 2023). Researchers have found that, with respect to AI adoption, highly educated workers might adapt easily or become self-employed compared to less educated or uneducated (Egana-del Sol et al., 2022; Fossen & Sorgner, 2022). Researchers have also highlighted the high demand for skills for AI adoption, such as digital literacy, analytical skills, complex cognitive skills (Jaiswal et al., 2022), machine learning (Fossen & Sorgner, 2022), and engineering analysis (Acemoglu et al., 2022). Consequently, the demand for highly skilled workers might increase by substituting mid-to low-skilled workers (Fossen & Sorgner, 2022). In addition, job postings would increase in technology-major firms (Acemoglu et al., 2022). Holm and Lorenz (2022) highlight that depending on the way AI is applied, it can also influence whether it will facilitate high-performing work by enhancing worker skills or decreasing autonomy by reducing learning opportunities and constraining skill usage. It also enhances job experience by providing better-quality data, insights, and more relevant knowledge resources (Bankins & Formora, 2023). Brivio et al. (2018) and Ocampo et al. (2022) demonstrated how AI can facilitate socialization in the workplace by conducting customized onboarding and offer AI chatbots, which can further support people like newcomers to handle stress. Besides, Gen-AI can transform tasks from complex to simple; be creative or personalize work processes, and these practices by fundamentally changing job titles, the skill required for those roles, and the recruitment criteria for talent (Fossen & Sorgner, 2022). Employees can also be seen changing their professional identities and expectations because their role now becomes more collaborative with AI systems (Bankins et al., 2023). On the other hand, leadership ideas regarding the future of work concern balancing technology with human-centric values, ethical use, and driving innovation. This refined scope underscores the study's practical implications by addressing the nuanced interplay of generative AI and HR decisions, offering insights into actionable strategies for organizations.

A large amount of research is also available on themes such as human-AI interaction, suggesting various elements that ease or hinder the process (Marikyan et al., 2022). For example, Gen-AI and sustainable HRM (Dey et al., 2022; Ren et al., 2023) and AI and entrepreneurial discourse (Jeremiah, 2025). Past research also highlights the differences, benefits, and limitations of traditional human roles (Candrian & Scherer, 2022; Pethig & Kroenung, 2023), their impact on job growth and skill demand (Egana-del Sol et al., 2022; Fossen & Sorgner, 2022), and their impact on HRM (Guest et al., 2022, Pereira et al., 2023). In the absence of AI the traditional human roles were limited to operational activities and were unable to focus much on employee well-being and inclusive practices for diverse workforce. Innocenti and Golin (2022) and Suseno et al. (2022) discussed fear-based employee response to AI using technology acceptance models. It has also been linked to the deskilling of professional work and unethical decision-making (Xue et al., 2022). Research on AI- human collaboration has also shown employee responses to gig work (Duggan et al., 2022) by exploring how AI-human collaboration shapes employee perceptions, adaptability, and challenges in gig work settings. Dwivedi et al. (2023) discussed the implications of human-AI collaboration in educating and training workers by creating adaptive learning environments and personalized training content. The AI and human collaboration can bridge knowledge gaps and make the employees more AI driven future ready. Among other human applications, AI has been found to be useful in disciplines such as psychology and counselling to promote health and well-being (Pataranutaporn et al., 2021). The existing literature also highlights the potential of AI-based machine learning in diverse HR activities (Edwards et al., 2024; Malik et al., 2022), such as diversity inclusion (Daugherty et al., 2018), facilitating recruitment activities (Pan et al., 2022). AI also supports HR analytics (Margherita, 2022). The ability of AI to handle large amounts of data also shows its usage in the performance evaluation function by creating a clear link between employee performance and organizational-level outcomes (DeNisi et al., 2021).

According to a report by McKinsey, 'the economic potential of gen-AI (2023)' is approximately 75% of the gen-AI potential value in four areas: customer operations, marketing and sales, software engineering and R&D (McKinsey Digital, 2023). For instance, customer service agents are assisted by AI tools that help them better resolve client challenges by offering real time insights, recommendations and auto-generated responses. AI services can also create marketing content tailored to the micro-niches of potential customers by analyzing vast amount of data related to customer demographics, behavior and preferences. Researchers use AI tools to quickly generate many customized drafts and designs using machine learning algorithms for market segmentation. AI adoption alters the organizational hierarchical

structure by flattening it, increasing the share of work at the junior level, and decreasing the work at middle and senior roles (Brynjolfsson et al., 2023). By revolutionizing internal knowledge management systems, the NLP ability of gen-AI can support staff in retrieving information efficiently by formulating natural and conversational questions akin to interacting with humans. In 2012, MGI estimated that employees spend one-fifth of their time searching for and gathering information, which is now saved with Gen-AI. Besides the available work, in the past few decades, there has been a significant increase in researchers' inclination toward human and AI collaboration (Budhawar et al., 2023; Chowdhury et al., 2023).

A review of the literature above shows that human AI collaboration is becoming common in workplaces to improve efficiency and job experience (Malik et al., 2022). This has led to discussions about the impact of AI on job roles, what employees are responsible for, and their sense of purpose at work (Wilson et al., 2017). However, considering the drawbacks and unforeseen challenges, the application of AI requires certain modifications in existing human job roles (Chowdhury et al., 2022; Kelan, 2024) to avoid the 'so-called mismanagement of information and technology' (Brynyolfsson, 1993). Since AI is replacing jobs that can be automated, freeing up employees from repetitive and routine work, and sparing their attention on more important complex and high-level jobs, it clearly requires certain modifications in the existing job roles. Workers now need to apply creativity and intelligence and adapt to more complex tasks. It is also expected that traditional job titles, such as lawyers, engineers, and doctors, would become uncommon. Instead, we might see more workers with diverse skills working in various organizations and industries. These individuals would not be limited to specific fields but would be highly acceptable and able to apply their critical thinking and innovation abilities to a range of roles.

Gaps in existing literature

The present research recognizes that generative AI can revolutionize workplaces. Research is available that highlights the integration of AI in business strategies. However, there is a lack of knowledge regarding the specific decisions required to optimize employee roles during integration. The selected body of literature also represents a balanced view of theory and practice and pinpoints gap in the absence of formal prioritization frameworks for better integration between humans and AI. This glaring gap formed the basis for a solid foundation using MCDMs. To fill this gap, we used the AHP method. We prioritize the criteria and alternatives for decisions related to value co-creation.

Application of resource-based view to the study

In the present research, a resource-based view (RBV) offers a strong theoretical foundation for understanding how companies can strategically employ and leverage human and technical resources to optimize employee roles in the context of gen-AI (Boon et al., 2018; Paauwe & Boselie, 2003). It serves as a lens for researchers to explore the advantages of integrating gen-AI with the workforce. For instance, Chaudhary (2023) highlights improvements in HR processes, such as more cost effective and better data management, easy recruitment and onboarding. The notion of RBV leads to an examination of how companies allocate human and technical resources to optimize employee roles (Boon et al., 2018). It also investigates how investing in AI digital technology and relevant employee training will lead to co-creation (Paauwe & Boselie, 2003; Susskind and Susskind, 2022). The RBV acknowledges the significance of dynamic capabilities for organizations to adapt to changing environments (Wade and Hulland, 2004). The RBV suggests that resources should complement each other to gain a competitive advantage through co-creation, knowledge sharing, and knowledge circulation between organizations and individuals (Mikalef and Gupta, 2019; Renkema, 2022). In this study, based on the RBV theory, we examine the synergistic outcomes from human skills and gen-AI and optimize value co-creation in employee roles.

By delving into the unexplored realm of the integration of gen-AI into the workplace, we aim to provide valuable knowledge for researchers and businesses. We expect that the findings of this study will guide organizations in developing strategies for human AI collaboration and ensure that employee roles evolve and improve in the digital era.

Research objectives

The key objective of the research study is to identify and compare the dimensions that should be considered when rethinking employee roles to facilitate effective value co-creation. It can also be useful to explore and analyze the alternatives that companies could pursue to maximize employee involvement within the framework of value co-creation and generative AI.

Research methodology

The aim of this study is to identify the most important criteria that must be considered while rethinking employee roles for the effective integration of AI into tasks. The study also attempts to measure and contrast various decision options that companies could pursue to maximize employee involvement within the framework of value co-creation with generative AI. To achieve these objectives Systematic literature review and MCDMs can be appropriate. Systematic literature review can be insightful to achieve identify and compare the dimensions that should be considered when rethinking employee roles to facilitate effective value co-creation. It can also be useful to explore and analyze the alternatives that companies could pursue to maximize employee involvement within the framework of value co-creation and generative AI. MCDMs can be insightful to measure and contrast the influence of various decision options as they are designed to manage situations with numerous competing decision alternatives (Marques, 2020). It offers a structured method to assess various options and select the best course of action in research. In comparison to other MCDMs, AHP stands out as a better choice because of the intricate interplay between human intelligence and AI capabilities and value co-creation dynamics; the hierarchical nature of AHP allows detailed analysis of decision points, enabling researchers to capture the intricate relationship between various criteria and alternatives (Agarwal and Ojha, 2024). It provides a flexible way to include both the qualitative and quantitative criteria essential for the analysis. AHP allows for a systematic analysis of factors such as productivity, creativity, adaptability and ethical considerations, providing valuable insights into the most effective allocation of roles within organizations navigating the integration of AI technologies. In addition, AHP offers a robust method for handling inconsistencies and uncertainties in decision-making. By employing pairwise comparisons and sensitivity analysis, researchers can validate the robustness of their findings and enhance the credibility and reliability of research outcomes. The following are the steps involved in the application of AHP to optimize employee roles while adopting gen-AI (see Figure 1):

Defining the goal

First, the main goal of the decision-making process must be determined. The goal is to optimize employee roles for human and AI integration for value co-creation.

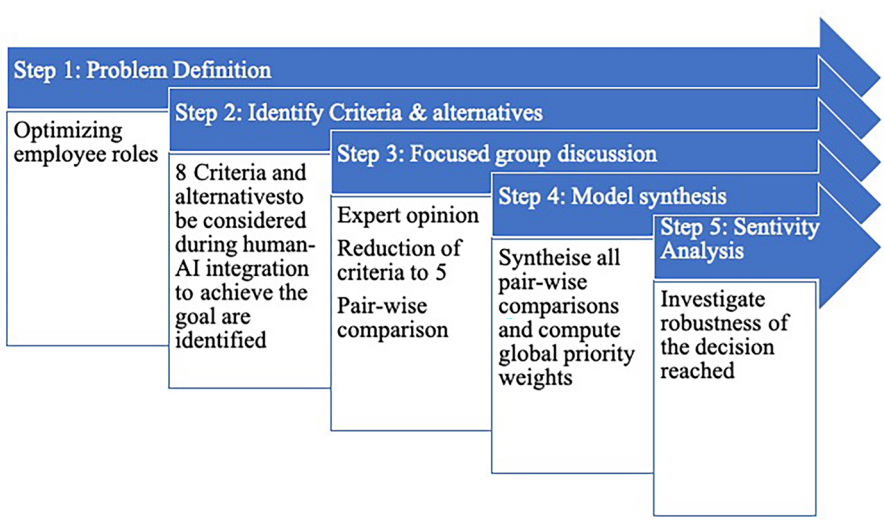

Figure 1. Research design (author's own).

Identification of critical decision criteria and alternatives for human and AI integration following systematic literature review

Next, a hierarchal structure is created with goals, criteria, and alternatives. For this purpose, criteria and alternatives were explored by undertaking a transparent, replicable, and rigorous literature review. The operation starts with the specification of keywords-defined, targeting keywords for example 'human-AI integration,' 'decision-making criteria,' 'value co-creation dynamixs,' 'AI in workflows' and 'organizational adaptability.' These keywords were run on following databases: Scopus, Web of Science, and IEEE Xplore. Only peer-reviewed published articles within 10 years of the study were considered including organizational AI adoption, decision-making frameworks, and optimization of roles but excluding empirical or organizationally irrelevant papers. In the primary set of 500 studies, about 60 articles were refined on the basis of quality and relevance. Analysis of these articles lead to major criteria for decision making like enhancing productivity, creating avenues for innovation, being ethical, developing employee skills and data security after being found prevalent and important in the literature. In all eight decision-making criteria were identified (Table 1). Besides, the theme wise analysis if the papers led to decision alternatives, for instance, task automation, hybrid roles, continuous learning programs, and cross-functional collaboration platforms are derived from the action-practice and strategic focus in the studies reviewed. The decision-making criteria depict the dimensions that organizations must consider when redefining employee roles during Gen-AI implementation for effective value co-creation.

Decision alternatives involve various strategies, approaches, and configurations for integrating generative AI into the workforce. The twelve possible decision alternatives are listed in Table 2. They are the strategic directions that organizations might take to maximize employees' involvement within the framework of gen-AI.

The 12 decision alternatives influence the criteria to varying degrees; some have a significant influence, while others have a weak influence. A few may have an indirect or negative influence. Therefore,

Table 1. Key decision criteria for human and AI integration.

	Decision-criteria	Definition	Source
1.	Productivity enhancement (PE)	The degree to which, the redefined employee roles contribute to increased productivity by seamlessly incorporating advanced AI technology to streamline processes, automate everyday tasks, anticipate future needs and provide tailored suggestions.	Noy and Zhang, 2023; Baily et al., 2023
2.	Innovation (INN)	Whether the new employee roles have the capability to promote innovation by using generative AI for creative problem-solving, idea generation, prototype development and product development.	Brynyolfsson et al., 2023; Su & Yang 2023
3.	Ethical considerations (EC)	The ethical implications of revised employee roles, such as transparency, bias, fairness, accountability, trust and responsible use of AI.	Siau and Wang, 2020
4.	Employee skill development (ESD)	If it enables the growth and improvement of employee skills to effectively work alongside gen-AI, guaranteeing comfortable transition and ongoing adjustment.	Dwivedi et al., 2023; Gozalo-Brizuela and Garrido-Merchan, 2023; Su and Yang, 2023
5.	Change in customer experience (CCE)	How redefined employee roles can enhance the customer experience, satisfaction, and engagement by utilizing gen-AI during customer interactions?	Malik et al., 2022
6.	Organizational adaptability (OA)	The organization's ability to adjust to changes in employee roles depending on factors like organizational culture, change management, structural adjustments, leader support, colleagues' attitudes towards change, and employee morale.	Brynjolfsson and Mcafee (2017)
7.	Cost-benefit analysis (CBA)	The financial impacts of changing roles, such as training costs, acquiring data, computational resources, setting up gen-AI systems, building infrastructure, and the potential for returns on investment. Even though the costs can be high, sometimes the benefits outweigh the expenses.	Malik et al., 2022
8.	Data security and privacy (DSP)	While incorporating gen-AI into employee tasks, data security and privacy remain the priority. This is crucial in order to prevent data breaches, protect sensitive information, and uphold privacy policies.	Siau and Wang, 2020; Fang et al., 2017

Source: author's own.

careful investigation of the decision criteria regarding decision-making alternatives is required. A pairwise comparison was conducted for this purpose.

Pairwise comparison

Subsequently, experts and stakeholders were approached to compare the criteria and alternatives in pairs by judging their relative importance. Seven experts from academia and corporations were approached to understand the research dimensions and their dynamics. In this study, the focus group method was utilized to gain insights into optimizing employee roles in the gen-AI era. This method is preferred owing to its ability to thoroughly explore expert opinions and perspectives on the comparison between criteria for job effectiveness and decision alternatives for the effective implementation of AI and human collaboration. The focus group approach fosters conversational interactions and facilitates the generation of substantial qualitative data (Halliday et al., 2021). In addition, it allows researchers to delve into participants' experiences, attitudes, and preferences within time and financial constraints (Malhotra and Dash, 2020). The demographic information of the seven domain experts who contributed to this research is presented in Table 3. The selected experts possessed extensive knowledge of human resource management, AI, machine learning, and collaborative dynamics.

Before beginning the process of pairwise comparisons, the experts were shown the criteria and alternatives for the effective integration of AI into workflows. The aim of this exercise is to (i) take all experts on common definitions of the variables, and (ii) confirm the appropriate selection of criteria and alternatives. After a detailed discussion, consensus was reached on all alternatives, while the three criteria were merged with others due to their similar nature. For instance, productivity enhancement (PE) is merged with cost-benefit analysis (CBA), innovation (INN) with employee skill development (ESD), and data security and privacy (DSP) with ethical considerations (EC). The final set of criteria and alternatives is depicted in hierarchical form in Figure 2.

Thereafter, a pairwise comparison of the five revised AI adoption criteria with respect to (w.r.t) the goal of optimizing employee roles is first performed. Then, pairwise comparisons of each decision alternative corresponding to each criterion were performed separately. The reason for separately determining the relative weights of the decision alternatives is the varying importance of the same alternative for different criteria. For instance, INN and CBE might be more important for EEP than for TRV. A questionnaire consisting of all elements of the AHP hierarchy was created to collect responses from the experts. A comparison between the criteria and alternatives was performed using Saaty's 9-point rating scale (see Table 4).

Model synthesis

The pairwise comparisons performed in the previous step are then synthesized through the calculation of priority weights using mathematical algorithms such as the eigenvector method (Wan et al., 2021). Version 3.2.0 of super decision software is utilized for the synthesis of all pair-wise comparisons wherein judgement of the experts and ranks and importance of each criterion and alternatives are determined (Sharma et al., 2021). Model synthesis also involves normalizing the local priority weights of all the alternatives. They were combined with respect to each criterion to obtain the global composite priority weights of all decision alternatives shown at the third level of the AHP model. The consistency ratio (CR) for each set of comparisons was also computed. As a rule, a value below 1 is acceptable and indicates consistency in assigning pairwise judgements by expert respondents (Saaty, 1990).

Results

Systematic review of the literature resulted in the most important dimensions in the form of eight criteria among which the eight criteria were merged into five due to similar definitions during the focused group discussion. These dimensions must be taken into account by firms when rethinking employee

Table 2. Key decision alternatives for human and AI integration.

Ta	Decision-criteria	Definition	Source
1.	Task automation (TA)	Transform the way employees work by using cutting-edge AI technology to streamline mundane and repetitive tasks. This empowers employees to dedicate their time and energy to the more challenging and innovative aspects of their roles.	Jackson et al., 2024
2.	Augmentation and Collaboration (AC)	Foster teamwork among staff members and leverage creative AI technology to boost human skills, improving productivity and fostering innovation.	Siemon, 2022
3.	Specialized AI teams (ST)	Establish specialized teams that concentrate on building and overseeing innovative AI systems, enabling other staff members to utilize AI knowledge without needing to directly participate in AI-specific duties.	Susskind & Susskind, 2022
4.	Continuous learning and development (CLD)	Create ongoing learning initiatives and training opportunities to empower employees with the skills and knowledge needed to collaborate effectively with AI technology.	Fast and Horvitz, 2017; Tcharnetsky and Vogt, 2023)
5.	Hybrid roles (HR)	Update job descriptions to include hybrid roles that combine traditional duties with tasks that utilize gen-AI technology.	Perifanis & Kitsios, 2023
6.	Ethics oversight and governance (EOG)	Implement specific. positions dedicated to supervising the ethical utilization of gen-AI within the company, guaranteeing compliance with ethical guidelines and protocols.	de Almeida, 2021
7.	Customer interaction enhancement (CIE)	Improve employee roles to boost customer engagement through the use of advanced AI technology for more tailored and effective customer support.	Malik et al., 2022
8.	Innovation labs (IL)	Create innovation labs or designated areas within the company where staff can work together with gen-AI to discover and create creative solutions.	Brynyolfsson et al., 2023; Su & Yang 2023
9.	Adaptive organizational structure (AOS)	Make changes to the company's structure to incorporate gen-AI, so that the decision-making and reporting systems match the new changes.	Brynjolfsson and Mcafee (2017)
10.	Task redesign for value co-creation (TRV)	Revamp tasks and duties to concentrate on value co-creation by utilizing gen-AI to enhance creative and collaborative endeavors.	Xu et al. 2023; Fugener et al., 2021
11.	Employee empowerment programs (EEP)	Develop programs that enables employees to play an active role in improving and optimizing gen-AI systems utilized in their work.	Fast and Horvitz, 2017; Varma et al., 2023
12.	Cross-functional empowerment programs (CFEP)	Create systems that encourage teamwork across departments, enabling employees to work together using advanced AI technology on projects that involve multiple disciplines.	De Vos., 2020; Tcharnetsky and Vogt, 2023)

Source: author's own.

roles to facilitate effective value co-creation. Table 7 shows the outcomes of the pair-wise comparison syntheses. The results of the AHP application are shown in Tables 5 – 7. The inconsistency statistics of all comparisons were below the threshold limit of 0.1 and are depicted in the respective tables. Further, the result of the first set of pairwise comparisons, that is, comparison of criteria with respect to the goal, shows three dominating criteria that support the goal of optimizing employee roles in AI integration. The results show high weights of innovativeness (38.1%), productivity enhancement (23.3%) and change in customer experience (20.8%). Also, the literature review resulted in 12 alternatives that companies could pursue to maximize employee involvement within the framework of value co-creation and generative AI. Moreover, the results of the pairwise comparisons of each alternative corresponding to all criteria are presented in Table 6. It shows the relative weight of each alternative for AI collaboration with respect to individual AI integration criteria. In addition, Table 7 shows the outcomes from the model synthesis, that is, the overall rank based on globalized weights of alternatives with respect to the goal and criteria. Adaptive organizational structure (AOS) (16.4%), specialized AI teams (ST) (12.7%), ethics oversight and governance (EOG) (12%), innovation labs (IL) (10.2%) and continuous learning and development (CLD) (9.8%) held the top five ranks for optimizing employee roles during human AI integration.

Discussions

The results of the pairwise comparison of decision criteria seem meaningful for the manufacturing industry, as human-AI integration must bring innovativeness to transform the processes by automating

Table 3. Demography of the experts referred to in the focused group approach.

S. No	Educational qualifications	Designation, domain expertise and duration of experience	Industry	Insights
1.	B. Tech (Computer Science)	AI research scientist, 15 years of experience in Artificial Intelligence, machine learning, natural language processing	Information technology	Technical aspect of integrating AI into organizational workflows
2.	B. Tech (Computer Science), MBA (Finance)	Business Analyst, 15 years of experience in cost-benefit analysis, financial planning	Fintech	Insights into economic factors influencing decision-making
3.	B. Tech (Information Technology)	IT specialist (Information security), 25 years of experience in cybersecurity, data security & privacy	Brokerage firm	Examine risks associated with data breach, privacy in AI integration. Recommendations on data safety and cyber security compliance
4.	MBA (HR)	HR Specialist, 15 years of experience in workforce management, employee training, job design	Manufacturing	Understanding employee roles while AI integration
5.	MBA (HR), Diploma in Industrial relations management	HR Manager, 20 years of experience in Legal compliance	Manufacturing	Guidance in ensuring ethical considerations
6.	PhD	Professor, 40 years of experience in teaching and research (HR & OB)	Education and research	Behavioral aspects of AI integration
7.	PhD	Professor, 25 years of experience in teaching and research (AI & ML)	Education and research	Contribution in AI ethics, implications of AI implementation

Source: author's own.

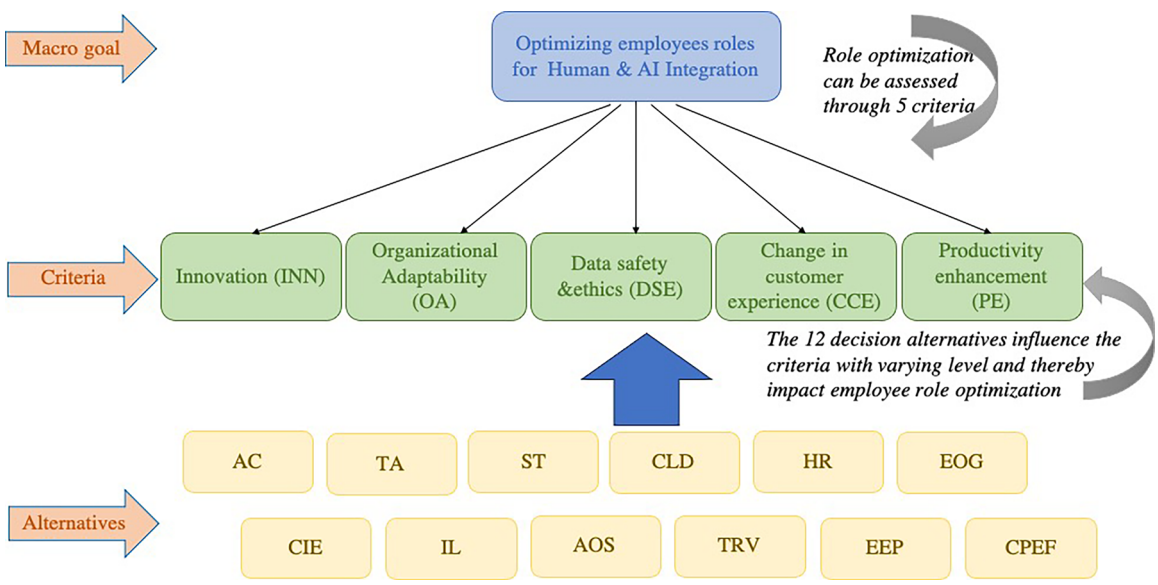

Figure 2. Hierarchy of elements for AHP (author's own).

Table 4. Saaty's scale of relative importance.

Intensity of importance	Definition
1	Equal importance
3	Moderate Importance
5	Strong importance
7	Very strong importance
9	Extremely strong importance

Source: author's own.

repetitive tasks, need-based tailored content, operational efficiency and quick adaptation to dynamic market trends (Eller, 2023). In addition, productivity enhancement can be justified as a crucial criterion because of the persistent challenge of achieving promised business gains from AI integration (Economist, 2020). Collaboration must bring a return on huge investments involved in implementing AI, data

infrastructure, redesign and reconfiguration of existing practices. The third important criterion for optimizing human roles during integration is that collaboration must exhibit the potential to enhance customer experience by offering quick and automated answers and insights. Otherwise, it could lead to frustration if it fails to maintain customer expectations regarding accuracy and responsiveness (Oxford Analytica, 2023).

Besides, the results of decision alternative comparison also seem logical and in sync with the available studies. For instance, adaptive organizational holding top rank is logical, as merely inventing new technology is not sufficient, but leveraging them effectively with appropriate support is necessary. Thus, AI integration into HR requires organizations to create the necessary conditions and introduce management practices (Brynjolfsson and McAfree, 2017). They must alter their work dynamics and reshape the cultural fabric of an organization. Moreover, having specialized AI teams is crucial for building a framework accommodating the Gen-AI strategy, handling resistance issues, other concerns, developing a safe environment for experimentation and learning, and implementing guidelines and governance for Gen-AI usage. Working with a specialized AI team can be insightful for HR staff from an engineering perspective (Gmeiner et al., 2023). Their role is to intervene, moderate, and provide input whenever needed (Siemon, 2022). The results of AHP model synthesis also highlight ethics oversight and governance as a significant alternative in AI integration, as incorporating regulatory measures to ensure responsible and ethical use of Gen-AI is a must. This ensures that the proposed framework accommodates the deployment and adoption of these regulatory frameworks. In addition, innovation laboratories are critical to actualizing the researcher's mindset by deeply understanding and experimenting with Gen-AI. In addition, owing to the specific requirements of Gen-AI integration, human resources must be empowered to increase productivity and stay ahead in dynamic and competitive marketplaces by imparting high technology skills. This justifies the fifth position of continuous learning and development in AI-human integration.

Sensitivity analysis

Sensitivity Analysis (SA) was conducted to investigate the robustness of the decision achieved if the criteria weight changes. SA is significant for understanding the stability of decision outcomes and the critical variables that impact the final decision. The changes were made in two phases. First, the criteria innovativeness (INN) is increased by approximately 25% (from 0.38 to 0.48). The results revealed that changes in the first criterion had no significant influence on alternative ranking. Thus, the overall rank of the final outcome remained unchanged in comparison with the original ranking, as shown in Table 7. Second, the criteria 'Productivity enhancement from investment made (PEI)' is increased by approximately 25% (from 0.233 to 0.291). The results showed that this criterion had no major impact on the final outcome. Similarly, the model is also retested for the change in outcome with 25% increase in the third criteria 'Change in customer experience (CCE) (from 0.208 to 0.26). Consistent with previous findings, this change did not have a substantial impact on the rankings (see Figure 3). This indicates that the model is stable, robust, and appropriate for decision making.

Theoretical and practical implications

This study aims to explore the criteria and alternatives for optimizing human roles while integrating AI with human work. The results reveal three important criteria that must be considered during human-AI collaboration. These are innovativeness (39.2%), productivity enhancement (20.6%) and

Table 5. Priority weights of the criteria for integration of AI in the workflows w.r.t goal i.e. optimizing employee roles.

Criteria	Normalized score
Innovativeness (INN)	0.381
Productivity enhancement from investment made (PEI)	0.233
Change in customer experience (CCE)	0.208
Data safety and ethics (DSE)	0.079
Organizational Adaptability (OA)	0.096
Inconsistency: 0.28362	

Source: author's own.

Table 6. Priority weights of the decision alternatives w.r.t criteria for integration of AI in the workflows.

	Organizational adaptability (OA)	Data safety and ethics (DSE)	Productivity enhancement from investment made (PEI)	Change in customer experience (CCE)	Innovativeness (INN)
Augmentation and Collaboration (AC)	0.062	0.214	0.0641	0.200	0.221
Adaptive organizational structure (AOS)	0.106	0.013	0.030	0.060	0.021
Customer interaction enhancement (CIE)	0.057	0.014	0.016	0.0539	0.009
Continuous learning and development (CLD)	0.036	0.036	0.062	0.028	0.011
Cross-functional empowerment programs (CFEP)	0.019	0.053	0.066	0.036	0.050
Employee empowerment programs (EEP)	0.202	0.028	0.070	0.053	0.185
Ethics oversight and governance (EOG)	0.114	0.029	0.090	0.160	0.154
Hybrid roles (HR)	0.100	0.041	0.111	0.135	0.079
Innovation labs (IL)	0.088	0.077	0.219	0.982	0.040
Specialized AI teams (ST)	0.024	0.114	0.158	0.046	0.48
Task automation (TA)	0.082	0.161	0.068	0.036	0.074
Task redesign for value co-creation (TRV)	0.105	0.214	0.040	0.090	0.103
Inconsistency	*0.65*	*0.375*	*0.61*	*0.507*	*0.43*

Source: author's own.

Table 7. Overall ranks of the decision alternatives from model synthesis.

Alternative	Priority weight	Rank
Task automation (TA)	0.032	11
Augmentation and Collaboration (AC)	0.025	12
Specialized AI teams (ST)	0.127	2
Continuous learning and development (CLD)	0.098	5
Hybrid roles (HR)	0.039	10
Ethics oversight and governance (EOG)	0.120	3
Customer interaction enhancement (CIE)	0.073	8
Innovation labs (IL)	0.102	4
Adaptive organizational structure (AOS)	0.164	1
Task redesign for value co-creation (TRV)	0.095	6
Employee empowerment programs (EEP)	0.077	7
Cross-functional empowerment programs (CFEP)	0.048	9

Source: author's own.

change in customer experience (20.5%). In addition, five key alternatives were discovered for employee role optimization for AI integration. It is expected that by understanding specific criteria that should be considered or ignored, organizational leaders can devise strategies to navigate the optimization of employee roles for the effective integration of AI into workflows. Furthermore, alternatives with high-priority weights will help executives prepare for a Gen-AI future, readying their workforce for automation, strategizing for augmentation, appreciating human-centric skills, and even pioneering new roles. The key theoretical contribution of this study involves exploring human resource dimensions of Gen-AI implementation. In addition, this study is among the few to apply MCDM (AHP) to the analysis of the dimensions. In fact, the application of the resource-based view in this study justifies the discussion against the backdrop of technology adoption and readiness at the organizational level.

Conclusions, limitations and future research

This study investigates the dynamics of employee roles and value co-creation within the context of Generative AI. It applies a rigorous multi-criteria decision-making (AHP) framework. The results indicate that AI and human collaboration decisions must be based on expected innovativeness, productivity enhancement and changes in customer experience. In addition, the AHP results also show that considering the wide range of organizational necessities associated with AI adoption, adaptive organizational structure, specialized AI teams, ethics oversight and governance, innovation labs, and continuous learning and development are key alternatives for successful integration. This implies human–AI collaboration by

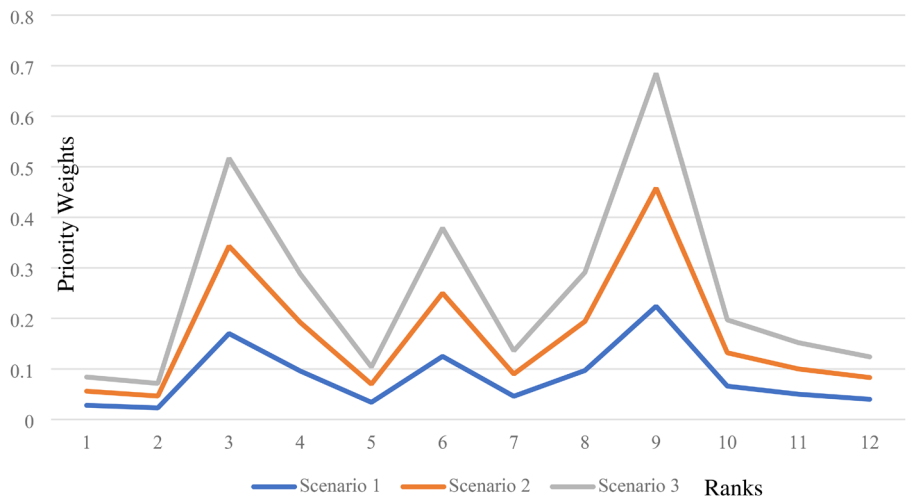

Figure 3. Results of sensitivity analysis (Source: author's own).

strategizing and fostering AI knowledge, research, and preparedness at the organizational level. However, this study used information from multiple sectors; therefore, the outcomes need to be validated for a specific sector. In addition, this study does not consider economic and business models that underpin human-AI collaboration. The influence of AI-human integration on revenue, cost structure, and business strategies has also not been discussed. The limitations of this study open avenues for future research.

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Data availability statement

The data that support the findings of the study are available from the corresponding author upon reasonable request.

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